Problem Summary

- Given the root of a binary tree and integer targetSum, determine if the tree has a root-to-leaf path such that adding all the node values along the path equals targetSum.

A root-to-leaf path means:

→ Starts at the root

→ ends at a leaf

→ a leaf has no left and right child

Key Idea

As we move down the tree, keep track of the current sum.

At each node:

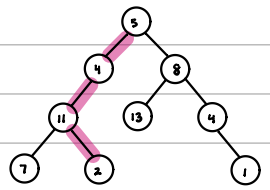
→ add the node's value to curr_sum

→ if we reach a leaf, check whether curr_sum == targetSum

→ Otherwise keep searching left or right

Diagram Examples

5+4+11+2 = 22



target Sum = 22

Approach

→ DFS Recursion

1) Start recursion from root with curr_sum = 0

2) If node is missing, return False

3) Add current node value to curr_sum

4) If the node is a leaf: return whether curr_sum == targetSum

5) Otherwise: recursively check left or right subtree

6) If either side returns True, return True

Time Complexity : $O(n)$ / $O(\log n)$ → Each node is visited once
(worst case / Balanced tree)

Space Complexity : $O(h)$ → Recursion stack, where h is tree height